

FastAPI — Complete Topic List

Agentic AI Roadmap ke liye — Kya Seekhna Hai

Note: Tumne Phase 1 mein FastAPI basics seekh li hai. Yeh list complete reference hai — roadmap ke hisaab se topics automatically cover honge.

1. FastAPI Basics (Phase 1 mein ho gaya) ■

- FastAPI app setup — `app = FastAPI()`
- Routes banana — `@app.get()`, `@app.post()`, `@app.put()`, `@app.delete()`
- Path parameters — `/items/{item_id}`
- Query parameters — `/items?q=search`
- Request body — Pydantic BaseModel se
- Response models — `response_model` parameter
- HTTP status codes — 200, 201, 400, 401, 404, 422, 500
- HTTPException — proper error responses
- Swagger UI — `/docs` pe automatic documentation
- uvicorn se server run karna

2. Pydantic + Validation (Phase 1 mein ho gaya) ■

- BaseModel se data schemas banana
- Field types — `str`, `int`, `float`, `bool`, `list`, `dict`
- Optional fields — `field | None = None`
- Default values set karna
- EmailStr — email validation
- Field validators — `@validator` decorator
- Nested models — model ke andar model

3. Database Integration (Phase 1 mein ho gaya) ■

- SQLAlchemy — ORM setup
- MySQL connection — `DATABASE_URL`
- Models banana — Column types
- Session management — `get_db` dependency
- CRUD operations — create, read, update, delete
- Alembic — database migrations (Phase 3 mein aayega)

4. Authentication (Phase 1 mein ho gaya) ■

- JWT tokens — access + refresh tokens
- bcrypt — password hashing
- OAuth2PasswordBearer — FastAPI auth scheme
- Dependencies — `Depends()` se auth inject karna

- Redis — token blacklist
- Rate limiting — slowapi ya custom middleware

5. Advanced FastAPI (Phase 8 mein aayega) ■

- Background Tasks — BackgroundTasks class
- Middleware — request/response intercept karna
- CORS — Cross-Origin Resource Sharing setup
- Dependency Injection — advanced Depends() patterns
- Lifespan events — startup aur shutdown handlers
- APIRouter — routes ko alag files mein organize karna
- Structured JSON logging — production ke liye
- Health check endpoints — /health, /ready

6. Async FastAPI (Phase 5-6 mein aayega) ■

- async def routes — async/await with FastAPI
- Async database — asyncpg ya databases library
- Async HTTP calls — httpx async client
- WebSockets — real-time communication
- Server-Sent Events (SSE) — streaming responses
- Async background tasks — asyncio.create_task()

7. Production FastAPI (Phase 8 mein aayega) ■

- Docker — FastAPI ko containerize karna
- Celery + Redis — background job queue
- ARQ — async Redis Queue (Celery ka async alternative)
- GitHub Actions CI/CD — test + deploy pipeline
- Environment variables — .env + production secrets
- Gunicorn + uvicorn — production server setup
- Deploy — Render, Railway, AWS EC2

8. AI-Specific FastAPI (Phase 3-6 mein aayega) ■

- Streaming responses — StreamingResponse for LLM output
- File upload — PDF/document ingestion endpoints
- Agent endpoints — LangGraph agent ko API se expose karna
- MCP server — FastAPI pe MCP protocol implement karna
- Cost tracking middleware — har request ka token cost log karna
- Async agent calls — non-blocking agent execution

Summary — Kab Kya Seekhna Hai

Phase	FastAPI Topics	Status
Phase 1	Basics, Routes, Pydantic, Auth, DB	■ Done
Phase 2	StreamingResponse for LLM	■ Done
Phase 3-4	File upload, RAG endpoints	■ Aayega
Phase 5-6	Async routes, Agent endpoints	■ Aayega
Phase 8	Background tasks, Docker, CI/CD	■ Aayega
Phase 10	MCP server on FastAPI	■ Aayega
Phase 11	Cost tracking middleware	■ Aayega

Yaad rakho: FastAPI ek tool hai — AI Engineering tumhara asli kaam hai. Roadmap follow karo, projects banao, aage badhte raho.